CROSSING CORES, GORENSTEIN MOMENT POLYTOPES, AND SHARP VOLUME GAPS FOR RANDOM TOEPLITZ AND HANKEL LAWS

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Abstract. The limiting spectral laws of random symmetric Toeplitz and Hankel matrices are known to exist, but, to our knowledge, no recognizable closed-form scalar identification, density formula, or Cauchy transform is known for either law. Their even moments are sums of rational polytope volumes indexed by pair partitions. We show that these polytopes have considerably more structure than the moment formula initially suggests. Repeated deletion of a cyclically adjacent pair produces a canonical crossing core, and every deletion splits off a lattice interval by an affine unimodular map. This gives an exact and invertible Catalan transform for volumes and, more generally, for Ehrhart polynomials.

For every parity-mixing pairing we construct an affine unimodular equivalence between its Hankel and Toeplitz moment polytopes. Every resulting moment polytope is a full-dimensional lattice polytope and is Gorenstein of index two; in particular, its Ehrhart numerator is palindromic and congruent modulo two to a binomial power. A hypercube support graph for a double-occurrence word is shown to be a cactus. Combining that classification with sharp concentration inequalities yields the higher-core Toeplitz gap

$$p_T(\rho) \le \frac{9}{20} \qquad (|\rho| \ge 4),$$

with equality only for the four-pair core represented by 12123434. We also classify all volume-1/2 cores: they are precisely the reduced three-pair Toeplitz cores, and only 123123 survives the Hankel parity restriction.

We compute both moment sequences exactly through order fourteen, including $m_8(\gamma_T) = 908/15$, and determine the first reduced Hankel layers exactly. An exact rank-four box-spline certificate proves the sharp all-order Hankel gap: the unique four-pair class has volume 11/30, while every larger reduced core has volume at most 1/3. Quantization of normalized volume yields explicit all-order moment improvements. We also derive a Poisson approximation for the complete peeling defect with sharp first-order total-variation constants. Finally, the moments rule out classical infinite divisibility for both laws and free infinite divisibility for the Hankel law. This supersedes [20].

1. Introduction

Let $(a_j)_{j\ge 0}$ be independent and identically distributed real random variables with mean zero, variance one, and finite moments of every order. Consider the symmetric Toeplitz and Hankel matrices

$$T_n = (a_{|i-j|})_{i,j=0}^{n-1}, \qquad H_n = (a_{i+j})_{i,j=0}^{n-1}.$$

Bryc, Dembo, and Jiang [6] proved that the empirical spectral distributions of $n^{-1/2}T_n$ and $n^{-1/2}H_n$ converge almost surely to deterministic symmetric probability measures γ_T and γ_H; Hammond and Miller [10] independently established the Toeplitz limit. The laws are universal under the usual hypotheses, have unbounded support, and are determined by their moments.

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The Toeplitz law has a bounded density [19], whereas the Hankel law is not unimodal [6, Prop. A.2]. To our knowledge, no recognizable closed-form scalar identification, density formula, or Cauchy transform is currently known for either law.

The moment method explains the difficulty. For Wigner matrices, only noncrossing pairings survive and each surviving pairing contributes one. For Toeplitz and Hankel matrices, crossing pairings survive as well, and a pairing contributes the Euclidean volume of a pairing-dependent polytope. At moment $2k$ there are $(2k-1)!!$ Toeplitz pairings and $k!$ parity-mixing Hankel pairings. Exact computation therefore appears, at first, to be intrinsically order by order.

The main purpose of this paper is to expose a stable geometric and combinatorial structure behind those sums. If a chord joins adjacent circular positions, deleting it splits a free lattice interval from the corresponding moment polytope. Repeated deletion is confluent and leaves a canonical reduced diagram, which we call the *crossing core* in this setting. In the terminology of [1, Defs. 3.1–3.2], the deleted chords are exactly the noncrossing, nondividing pairs, and [1, Thm. 3.14] already gives the inflation coefficient $\binom{2k}{k-r}$. What is new here is the canonical ear-reduction formulation and its affine-unimodular product factorisation, Ehrhart lift, and inverse moment transform.

The opening-coordinate description has two further consequences. First, for every parity-mixing pairing π, the Hankel polytope is integrally affinely equivalent to the Toeplitz polytope:

$$\mathcal{P}_H(\pi) \cong_{\mathbb{Z}} \mathcal{P}_T(\pi).$$

This equality is termwise, not merely an identity after summing over pairings. Second, the constraint matrix has the consecutive-ones property. It is therefore totally unimodular, so every moment polytope is a lattice polytope. Twice the polytope has a unique interior lattice point, and translation by that point is reflexive. The polytopes are consequently Gorenstein of index two.

The sharp volume estimates require more than total unimodularity. The sequence of open sets of a double-occurrence word is a closed walk in the discrete cube. Its support multigraph is a cactus whose blocks use disjoint coordinate sets. This gives an exact formula for the number of exceptional, or nonbasis, states and permits an all-order classification when only two such states occur. Sharp two- and three-slab integrations then prove

$$p_T(\rho) \le \frac{9}{20}$$

for every reduced Toeplitz core ρ with at least four pairs. Equality has one dihedral type, represented by 12123434. The same argument also proves that volume $1/2$ occurs exactly at the three-pair layer. In the parity-mixing class, only the fully crossing word 123123 remains.

The Hankel restriction is substantially more rigid. Square blocks cannot occur in its prefix-state cactus. There is a unique four-pair dihedral core, of volume $11/30$, while every reduced parity-mixing core with at least five pairs has at least four exceptional states. We prove the required four-slab concentration inequality in a stronger totally-unimodular form. Its Gale dual is an Eulerian, cosimple regular matroid of rank four; the rank-four classification reduces it to the K_5 and $M^*(K_{3,3})$ frames, and an exact rational certificate completes the finite step. Thus every larger reduced Hankel core has volume at most $1/3$. Exact rational enumeration gives the sharper layer maxima $1/3$, $23/90$, and $1/4$ at five, six, and seven pairs.

The structural results also strengthen the enumerative part of the paper. The Catalan transform lifts verbatim to lattice-point enumerators. To our knowledge, the integral-affine equivalence, index-two Gorenstein property, and resulting Ehrhart transform have not previously been recorded for these moment polytopes. The exact moment calculations through order

fourteen give the corrected value

$$m_8(\gamma_T) = \frac{908}{15},$$

in place of 964/15, and the primitive masses provide independent exact checks. These exact values first appeared in the author's computational preprint [20]; the present article supplies the core factorisation, geometric classification, and separate exact checks. The complete number of deleted pairs has a Poisson limit; a coupling argument gives total-variation error of order k^{-1}. The low moments also give several divisibility obstructions.

Bose and Vishwakarma [4, Thms. 4.3(i)–(ii), 4.4(i), 4.5(i), and Rems. 4.2–4.3] give scalar-valued noncommutative $*$-distributional representations of both limits. In their notation, γ_T is the law of $(Z_1 + Z_2)/\sqrt{2}$, where the Z_i have standard Gaussian marginals, while γ_H is the law of

$$\frac{Y_1 + Y_2}{\sqrt{2}},$$

where Y_1 and Y_2 are self-adjoint standard symmetrized-Rayleigh variables. The two joint dependences are specified through mixed-moment integrals, but do not match the standard independence notions considered there. The polytope identities developed here provide exact all-order scalar-moment checks for the normalized sums without determining either mixed $*$-law.

In the cited version 1, the denominator 2 printed in the displayed statement of Theorem 4.3(i) is a normalization typo: its proof, Remark 4.3, and the later normalized-sum statements use $\sqrt{2}$, which is the normalization used here.

Main results. For ease of reference, the principal conclusions are collected here.

(1) Ear deletion has a unique terminal core and gives an affine unimodular product

$$\mathcal{P}_E(\pi) \cong_{\mathbb{Z}} \mathcal{P}_E(\mathrm{core}(\pi)) \times [0,1]^{k-r}.$$

(2) The full moments and Ehrhart sums are exact Catalan transforms of their primitive-core counterparts, with explicit generating-function inverses.
(3) For every parity-mixing π, $\mathcal{P}_H(\pi) \cong_{\mathbb{Z}} \mathcal{P}_T(\pi)$.
(4) Every Toeplitz and admissible Hankel moment polytope is a full-dimensional lattice polytope, Gorenstein of index two, with $h^*_\pi(z) \equiv (1+z)^k \pmod 2$.
(5) Reduced Toeplitz cores with at least four pairs have volume at most 9/20, with equality only for the class of 12123434. Volume 1/2 occurs exactly for reduced three-pair Toeplitz cores; in the Hankel class it occurs only for 123123.
(6) The four-slab inequality holds for every totally unimodular matrix of four distinct noncoordinate, coefficient-sum-one rows. Consequently the unique reduced four-pair Hankel class has volume 11/30, and every reduced Hankel core with at least five pairs has volume at most 1/3.
(7) Normalized-volume quantization gives strict factorial-size improvements beyond the sharp Toeplitz extremal layer and yields the refined moment bounds in (9.4).
(8) The exact moments through order fourteen and the first three nontrivial reduced Hankel layers are determined; the complete peeling statistic has sharp first-order Poisson asymptotics; and the moments imply failure of classical infinite divisibility for both laws and failure of free infinite divisibility for γ_H.

2. Moment polytopes

Let $\mathcal{P}_2(2k)$ be the set of pairings of $\{0,1,\dots,2k-1\}$. Write

$$\pi = \{\{l_1,m_1\},\dots,\{l_k,m_k\}\}, \qquad l_c < m_c,$$

and let $v(s)$ be the pair containing slot s. Put $\sigma(s)=1$ at the first occurrence of a pair and $\sigma(s)=-1$ at the second. For $t=(t_1,\dots,t_k)$, define

$$S_\ell(t) = \sum_{s<\ell} \sigma(s) t_{v(s)},$$

and

$$\mathcal{P}_T(\pi) = \left\{(x_0,t)\in\mathbb{R}^{k+1} : 0 \le x_0 + S_\ell(t) \le 1,\ 0\le \ell < 2k\right\}.$$

We write $p_T(\pi) = \operatorname{vol}_{k+1}\mathcal{P}_T(\pi)$.

A pairing is *parity mixing* if each pair joins an even slot to an odd slot. For such a pairing, set

$$A_\ell(x_0,y) = (-1)^\ell x_0 + \sum_{s<\ell} (-1)^{\ell-1-s} y_{v(s)},$$

and define

$$\mathcal{P}_H(\pi) = \left\{(x_0,y)\in\mathbb{R}^{k+1} : 0 \le A_\ell(x_0,y) \le 1,\ 0\le \ell < 2k\right\}, \qquad p_H(\pi) = \operatorname{vol}_{k+1}\mathcal{P}_H(\pi).$$

Remark 2.1 (Empty pairing convention). For the zero-pair word we set

$$\mathcal{P}_T(\varnothing) = \mathcal{P}_H(\varnothing) = [0,1], \qquad p_T(\varnothing) = p_H(\varnothing) = 1.$$

Thus $L_{\mathcal{P}_E(\varnothing)}(N) = N+1$. With this convention, complete ear deletion of a noncrossing k-pair word gives the integral cube $[0,1]^{k+1}$.

Theorem 2.2 (Moment–volume formula [6, 10, 3])**.** *For every $k \ge 1$,*

$$m_{2k}(\gamma_T) = \sum_{\pi\in\mathcal{P}_2(2k)} p_T(\pi), \qquad m_{2k}(\gamma_H) = \sum_{\substack{\pi\in\mathcal{P}_2(2k)\\ \pi \text{ parity mixing}}} p_H(\pi).$$

All odd moments vanish.

The iid assumption in the opening paragraph is the clean standard hypothesis for the almost-sure limiting laws. Variants with independent triangular arrays require uniform moment and Lindeberg-type assumptions; mere entrywise finiteness of moments is not, by itself, an adequate replacement.

For a slot ℓ, let

$$O(\ell) = \{c : l_c < \ell \le m_c\}$$

be the set of pairs open immediately before that slot. The Toeplitz walk form is

$$W_\ell = x_0 + \sum_{c\in O(\ell)} t_c.$$

Definition 2.3 (Opening coordinates)**.** Label pairs by their first occurrence and put

$$\Phi_0 = W_0, \qquad \Phi_c = W_{l_c+1}, \quad 1\le c\le k.$$

The empty state and the states immediately following first occurrences are called *basis states*. Every other distinct open state is *exceptional*.

Lemma 2.4 (Opening basis)**.** *The map* $(x_0, t_1, \dots, t_k) \mapsto (\Phi_0, \dots, \Phi_k)$ *is unimodular. Every slot form has a unique representation*

$$W_\ell = \sum_{j=0}^{k} \alpha_{\ell j} \Phi_j, \qquad \alpha_{\ell j} \in \mathbb{Z}, \qquad \sum_{j=0}^{k} \alpha_{\ell j} = 1.$$

Two slot forms agree if and only if their open sets agree.

Proof. At the opening of pair c,

$$\Phi_c = x_0 + t_c + \sum_{\substack{d<c \\ d \in O(l_c)}} t_d.$$

In the variable order $(x_0, t_1, \dots, t_k)$, the matrix is lower triangular with unit diagonal. This proves unimodularity and integrality. Every Φ_j and every W_ℓ has x_0-coefficient one, which gives the coefficient-sum identity. The final assertion follows from algebraic independence of the t_c's in the open-set formula. □

Lemma 2.5 (Totally unimodular signed slot normals)**.** *The full matrix of slot forms in opening coordinates is totally unimodular. In particular, every slot form has all coefficients in* $\{0, \pm 1\}$*, and every exceptional centered slot normal has odd support of size at least three.*

Proof. Let A be the matrix of all slot forms in $(x_0, t_1, \dots, t_k)$-coordinates. Its x_0-column consists of ones, and the column of t_c is the indicator of the cyclic interval during which c is open. Thus A has the consecutive-ones property and is totally unimodular by the alternating-sign form of the Ghouila–Houri criterion [18].

Let B be the square submatrix formed by the $k+1$ basis-state rows. By Lemma 2.4, $\det B = 1$, and the matrix of all slot forms in opening coordinates is AB^{-1}. Pivoting a totally unimodular matrix on a nonsingular basis submatrix preserves total unimodularity, so AB^{-1} is totally unimodular. Its entries, being one-by-one minors, lie in $\{0, \pm 1\}$. The coefficient sum is one by Lemma 2.4; hence a noncoordinate row has $2s + 1 \geq 3$ nonzero entries for some $s \geq 1$. □

3. Canonical crossing cores

Definition 3.1 (Ear and core)**.** An *ear* is a pair whose endpoints are adjacent in cyclic order. Delete an ear by removing its two endpoints and retaining the induced cyclic order. A pairing is *reduced* if it has no ears.

Proposition 3.2 (Confluence)**.** *Repeated ear deletion terminates in a reduced pairing independent of the order of deletion, up to cyclic relabelling. We denote it by* $\mathrm{core}(\pi)$*.*

Proof. Termination is immediate. Two distinct ears have disjoint endpoints, and deleting either one preserves adjacency of the other. Thus the reduction is locally confluent. Induction on the number of pairs gives a unique terminal core. □

Proposition 3.3. *A pairing has empty core if and only if it is noncrossing. Hence empty-core pairings on* $2k$ *points are counted by*

$$\mathrm{Cat}_k = \frac{1}{k+1}\binom{2k}{k}.$$

Proof. Every noncrossing pairing has an ear, and deleting it preserves noncrossingness. Conversely, inserting an ear into a cyclic gap creates no crossing. □

Theorem 3.4 (Integral core factorisation)**.** *Let $E \in \{T, H\}$, with the parity-mixing restriction for H. If π' is obtained from π by deleting one ear, then*

$$\mathcal{P}_E(\pi) \cong_{\mathbb{Z}} \mathcal{P}_E(\pi') \times [0,1].$$

If π has k pairs and its core has r, then

$$\mathcal{P}_E(\pi) \cong_{\mathbb{Z}} \mathcal{P}_E(\mathrm{core}(\pi)) \times [0,1]^{k-r}.$$

Consequently, $p_E(\pi) = p_E(\mathrm{core}(\pi))$, and for the Ehrhart counting function $L_P(N) = \#(NP \cap \mathbb{Z}^{\dim P})$,

$$L_{\mathcal{P}_E(\pi)}(N) = (N+1)^{k-r} L_{\mathcal{P}_E(\mathrm{core}(\pi))}(N).$$

Proof. First note that a cyclic rotation is integral-unimodular, not merely volume preserving. In the Toeplitz case, take the new initial coordinate to be the old slot form W_j, and sign and permute the increment variables according to their new first occurrences. In the Hankel case, take the new initial coordinate to be A_j and permute the y-variables; its initial-coordinate coefficient is $(-1)^j$. Both changes have determinant ± 1.

We may therefore rotate the word so that the ear occupies consecutive slots $\ell, \ell+1$. In the Toeplitz walk, if t is its variable, then

$$W_{\ell+1} = W_\ell + t, \qquad W_{\ell+2} = W_\ell,$$

and t occurs nowhere else. Replacing t by $u = W_{\ell+1}$ has inverse $t = u - W_\ell$ and integral linear determinant one. The ear constraint is simply $0 \le u \le 1$, while all remaining constraints are those of π'.

For Hankel, $A_{j+1} = y_{v(j)} - A_j$. At an adjacent matched pair,

$$A_{\ell+1} = y - A_\ell, \qquad A_{\ell+2} = A_\ell.$$

The change $u = A_{\ell+1}$, with inverse $y = u + A_\ell$, gives the same product. Iteration proves the factorisation. Ehrhart counting is invariant under affine unimodular maps and multiplicative under Cartesian products. □

4. A termwise Hankel–Toeplitz equivalence

The parity restriction does more than select which pairings survive: on each surviving pairing it converts the Hankel polytope into the Toeplitz polytope.

Theorem 4.1 (Affine equivalence)**.** *For every parity-mixing pairing π,*

$$\mathcal{P}_H(\pi) \cong_{\mathbb{Z}} \mathcal{P}_T(\pi).$$

In particular, $p_H(\pi) = p_T(\pi)$, and the two polytopes have the same Ehrhart polynomial and h^*-polynomial.

Proof. Put $B_\ell = (-1)^\ell A_\ell$. If l_c is the first occurrence of pair c, then parity mixing gives

$$B_\ell = x_0 + \sum_{c \in O(\ell)} \varepsilon_c y_c, \qquad \varepsilon_c = -(-1)^{l_c}.$$

Let $\delta_\ell = \mathbf{1}_{\{\ell \text{ odd}\}}$. We claim

$$\delta_\ell = \sum_{c \in O(\ell)} (-1)^{l_c} = -\sum_{c \in O(\ell)} \varepsilon_c. \tag{4.1}$$

Both sides vanish at $\ell = 0$. At either endpoint s of a chord, both sides change by $(-1)^s$, because the two endpoints have opposite parity. This proves (4.1) by induction.

Now set

$$C_\ell = B_\ell + \delta_\ell = x_0 + \sum_{c \in O(\ell)} \varepsilon_c (y_c - 1).$$

The Hankel condition says $B_\ell \in [0,1]$ for even ℓ and $B_\ell \in [-1,0]$ for odd ℓ; equivalently, $0 \le C_\ell \le 1$ for every ℓ. Therefore

$$t_c = \varepsilon_c (y_c - 1)$$

is the required affine map. Its linear part is diagonal with entries ± 1, and its translation is integral. □

Corollary 4.2 (Termwise domination)**.** *For every $k \ge 1$,*

$$m_{2k}(\gamma_H) \le m_{2k}(\gamma_T),$$

with strict inequality for $k \ge 2$. More generally, for every $N \ge 0$,

$$\sum_{\substack{\pi \in \mathcal{P}_2(2k) \\ \pi \text{ parity mixing}}} L_{\mathcal{P}_H(\pi)}(N) \le \sum_{\pi \in \mathcal{P}_2(2k)} L_{\mathcal{P}_T(\pi)}(N).$$

Proof. Parity-mixing pairings form a subset of all pairings, and Theorem 4.1 identifies each selected Hankel summand with the corresponding Toeplitz summand, for volume and for Ehrhart counting. For $k \ge 2$ the subset is proper, while every omitted Toeplitz polytope is full dimensional and hence has positive volume. □

Proposition 4.3 (Even-degree crossing criterion)**.** *Let $G_\times(\pi)$ be the chord-intersection graph of a pairing: its vertices are the chords, and two vertices are adjacent when their endpoints alternate around the circle. Then π is parity mixing if and only if every vertex of $G_\times(\pi)$ has even degree. More generally, after retaining any subset of chords and suppressing all other endpoints, the induced subpairing is parity mixing if and only if every vertex of the corresponding induced subgraph of $G_\times(\pi)$ has even degree.*

Proof. Let a chord have endpoints $l < m$. Among the $m - l - 1$ endpoints strictly between them, every chord contained in that interval contributes two, every crossing chord contributes one, and every other chord contributes zero. Hence

$$\deg_{G_\times(\pi)}(c) \equiv m - l - 1 \pmod 2.$$

The endpoints l, m have opposite parity exactly when $m - l - 1$ is even. This proves the first assertion. Applying the same count after deleting the complementary set of chords proves the induced-subgraph statement. □

5. Lattice and Gorenstein geometry

Let A_π be the $2k$ by $(k+1)$ matrix whose ℓ-th row is the coefficient vector of W_ℓ in $(x_0, t_1, \dots, t_k)$. Thus

$$\mathcal{P}_T(\pi) = \{z : 0 \le A_\pi z \le \mathbf{1}\}.$$

Theorem 5.1 (Lattice and Gorenstein structure)**.** *For every pairing π, $\mathcal{P}_T(\pi)$ is a full-dimensional lattice polytope. It is Gorenstein of index two. More precisely, if*

$$m = (1, 0, \dots, 0) \in \mathbb{Z}^{k+1},$$

then m is the unique interior lattice point of $2\mathcal{P}_T(\pi)$, and

$$2\mathcal{P}_T(\pi) - m$$

is reflexive. The same conclusions hold for every parity-mixing Hankel polytope.

Proof. In cyclic slot order, the x_0-column of A_π consists entirely of ones, and the column corresponding to pair c is the indicator of the interval

$$l_c < \ell \le m_c.$$

Thus every column has the consecutive-ones property. Such a 0-1 matrix is totally unimodular: for any selected rows, give them alternating signs in their inherited order; the signed sum in the intersection with any interval column is 0 or ± 1, so the Ghouila–Houri criterion applies; see also [18]. The matrix has full column rank by Lemma 2.4; the point $(1/2, 0, \dots, 0)$ satisfies every inequality strictly, and the opening constraints make the polyhedron bounded. Hence $\mathcal{P}_T(\pi)$ is full dimensional. Total unimodularity and the integral right-hand side imply that all vertices are integral. Moreover, $\mathcal{P}_T(\pi)$ has no interior lattice point: for integral z, the vector $A_\pi z$ is integral and cannot lie strictly between zero and one in every coordinate. Thus the codegree is at least two.

The dilate $2\mathcal{P}_T(\pi)$ is given by $0 \le A_\pi z \le 2$. Since $A_\pi m = \mathbf{1}$, the point m is interior, so the codegree is exactly two. If an integral point z is interior, then every entry of the integral vector $A_\pi z$ lies strictly between zero and two and must therefore equal one. Full column rank forces $z = m$.

Finally,

$$2\mathcal{P}_T(\pi) - m = \{v : |a_\ell \cdot v| \le 1, 0 \le \ell < 2k\},$$

where a_ℓ are the integral rows of A_π. Its polar is

$$\operatorname{conv}\{\pm a_\ell : 0 \le \ell < 2k\},$$

which is a lattice polytope. Thus the translate is reflexive, proving the Gorenstein assertion. The Hankel statement follows from Theorem 4.1. □

Corollary 5.2. *If π has k pairs, the Ehrhart numerator $h^*_\pi(z)$ is palindromic of degree k, with constant and leading coefficients equal to one. Moreover,*

$$(k+1)!\, p_E(\pi) \in \mathbb{Z}.$$

Proof. The polytope has dimension $k+1$ and codegree two, so the standard Gorenstein reciprocity theorem [5] gives palindromicity and degree k. The final assertion is integrality of normalized lattice volume. □

Corollary 5.3 (Parity refinement)**.** *For every k-pair Toeplitz polytope, and for every admissible Hankel polytope,*

$$L_{\mathcal{P}_E(\pi)}(N) \equiv \begin{cases} 1 \pmod 2, & N \text{ even}, \\ 0 \pmod 2, & N \text{ odd}. \end{cases}$$

Coefficientwise over $\mathbb{F}_2$,

$$h^*_\pi(z) \equiv (1+z)^k. \tag{5.1}$$

If $s_2(k)$ denotes the number of ones in the binary expansion of k, then

$$(k+1)!\, p_E(\pi) = h^*_\pi(1) \ge 2^{s_2(k)}. \tag{5.2}$$

In particular, every normalized volume with $k \ge 1$ is even.

Proof. The involution $z \mapsto m - z$ preserves $\mathcal{P}_T(\pi)$, because $A_\pi m = \mathbf{1}$. On the lattice points of $N\mathcal{P}_T(\pi)$ it becomes $z \mapsto Nm - z$. It has no fixed point when N is odd and has the unique fixed point $Nm/2$ when N is even. The Hankel claim follows from Theorem 4.1.

Reducing the Ehrhart-series identity

$$\sum_{N\geq 0} L_{\mathcal{P}_E(\pi)}(N)z^N = \frac{h^*_\pi(z)}{(1-z)^{k+2}}$$

modulo two gives

$$\frac{1}{1-z^2} = \frac{h^*_\pi(z)}{(1-z)^{k+2}},$$

and hence (5.1). The number of odd coefficients of $(1+z)^k$ is $2^{s_2(k)}$. Since the coefficients of an h^*-polynomial are nonnegative integers and their sum is normalized volume, (5.2) follows. □

6. Catalan inflation and exact transforms

Let $\mathcal{R}_{T,r}$ be the reduced pairings on $2r$ cyclically labelled points, and let $\mathcal{R}_{H,r}$ be the reduced parity-mixing pairings. A function on reduced pairings is *rotation invariant* if cyclic relabelling does not change its value.

Theorem 6.1 (Catalan inflation)**.** *Let $E \in \{T, H\}$ and let F be rotation invariant. For $1 \leq r \leq k$,*

$$\sum_{\substack{\pi\in\mathcal{P}_E(2k)\\ |\operatorname{core}(\pi)|=r}} F(\operatorname{core}(\pi)) = \binom{2k}{k-r} \sum_{\rho\in\mathcal{R}_{E,r}} F(\rho), \tag{6.1}$$

where $\mathcal{P}_T(2k) = \mathcal{P}_2(2k)$ and $\mathcal{P}_H(2k)$ is the parity-mixing class.

Proof. Every deleted chord lies in one of the $2r$ gaps between consecutive surviving core vertices; otherwise it could not become an ear without deleting a core vertex. In each gap the deleted chords form a noncrossing pairing. Conversely, arbitrary noncrossing decorations in the gaps reduce to the prescribed core.

Let

$$C(z) = \sum_{j\geq 0} \operatorname{Cat}_j z^j = \frac{1-\sqrt{1-4z}}{2z}.$$

Count globally the decorated pairings together with a marked surviving core vertex. Marking after decoration gives $2r$ choices for each pairing on the left-hand side. In the other direction, choose the absolute label of the marked vertex in $2k$ ways, choose the reduced core on the standard labels, and choose its $2r$ noncrossing gap decorations. The latter decorations using $k-r$ pairs are counted by $[z^{k-r}]C(z)^{2r}$. Equating these two weighted counts gives the factor

$$\frac{k}{r}[z^{k-r}]C(z)^{2r}.$$

Lagrange inversion gives

$$[z^m]C(z)^{2r} = \frac{r}{m+r}\binom{2m+2r}{m},$$

which yields $\binom{2k}{k-r}$. This coefficient is also the partial-pairing count in [1, Theorem 3.14].

In the parity-mixing class each inserted gap contains an even number of vertices. Every chord in a noncrossing pairing on a consecutive even interval has endpoints of opposite parity. The same construction therefore remains within the parity-mixing class. □

Remark 6.2. The case $r = 1$ in Theorem 6.1 is vacuous because no one-pair word is reduced. The case $r = 0$ is separate: empty-core pairings are precisely the Cat_k noncrossing pairings, and every noncrossing pairing is parity mixing.

Define the primitive core masses

$$\kappa_r^E = \sum_{\rho \in \mathcal{R}_{E,r}} p_E(\rho), \qquad r \geq 2.$$

There is no reduced one-pair core, and there is no reduced parity-mixing two-pair core. By Corollary 4.2, $\kappa_r^H \leq \kappa_r^T$ for every $r \geq 2$.

Corollary 6.3 (Moment–core transform)**.** *For $E \in \{T, H\}$ and $k \geq 1$,*

$$m_{2k}(\gamma_E) = \mathrm{Cat}_k + \sum_{r=2}^{k} \binom{2k}{k-r} \kappa_r^E. \tag{6.2}$$

Proof. The empty-core contribution is Cat_k. For nonempty cores, Theorem 3.4 makes volume a function of the core, and Theorem 6.1 applies. □

Put

$$M_E(z) = \sum_{k \geq 0} m_{2k}(\gamma_E) z^k, \qquad K_E(u) = \sum_{r \geq 2} \kappa_r^E u^r, \qquad m_0 = 1.$$

Theorem 6.4 (Generating-function inverse)**.** *With $u = C(z) - 1$,*

$$M_E(z) = C(z) + \frac{K_E(u)}{\sqrt{1-4z}}. \tag{6.3}$$

Equivalently, since $z = u/(1+u)^2$,

$$K_E(u) = \frac{1-u}{1+u}\left[M_E\left(\frac{u}{(1+u)^2} \right) - (1+u) \right]. \tag{6.4}$$

For $r \geq 2$,

$$\kappa_r^E = \sum_{k=2}^{r} (-1)^{r-k} q_{r,r-k} \big(m_{2k}(\gamma_E) - \mathrm{Cat}_k \big), \tag{6.5}$$

where

$$q_{r,j} = \frac{2r}{2r-j} \binom{2r-j}{j}$$

is the number of j-edge matchings of the cycle C_{2r}.

Proof. The identity

$$\sum_{k \geq r} \binom{2k}{k-r} z^k = \frac{(C(z)-1)^r}{\sqrt{1-4z}}$$

turns (6.2) into (6.3). The relation $C = 1 + zC^2$ gives (6.4). Extracting coefficients of $(1-u)(1+u)^{-2k-1}$ gives

$$[u^j](1-u)(1+u)^{-2k-1} = (-1)^j \frac{2r}{2r-j} \binom{2r-j}{j}, \qquad r = k + j,$$

which is (6.5). □

The first primitive masses are

r	2	3	4	5	6	7
κ_r^T	$\frac{2}{3}$	2	$\frac{178}{15}$	$\frac{253}{3}$	$\frac{24798}{35}$	$\frac{206797}{30}$
κ_r^H	0	$\frac{1}{2}$	$\frac{11}{15}$	$\frac{131}{36}$	$\frac{83}{5}$	$\frac{63907}{720}$

(6.6)

Corollary 6.5 (All-order lower bounds)**.** *For $k \geq 3$,*

$$m_{2k}(\gamma_T) \geq \mathrm{Cat}_k + \frac{2}{3}\binom{2k}{k-2} + 2\binom{2k}{k-3},$$

and for $k \geq 4$,

$$m_{2k}(\gamma_H) \geq \mathrm{Cat}_k + \frac{1}{2}\binom{2k}{k-3} + \frac{11}{15}\binom{2k}{k-4}.$$

Proof. Retain in Corollary 6.3 only the nonnegative primitive contributions displayed in (6.6): $r = 2, 3$ for Toeplitz and $r = 3, 4$ for Hankel. □

The same transform holds before taking leading Ehrhart coefficients. Define

$$\mathsf{A}_k^E(N) = \sum_{\pi \in \mathcal{P}_E(2k)} L_{\mathcal{P}_E(\pi)}(N), \qquad \mathsf{K}_r^E(N) = \sum_{\rho \in \mathcal{R}_{E,r}} L_{\mathcal{P}_E(\rho)}(N),$$

and write $X = N + 1$.

Theorem 6.6 (Ehrhart core transform)**.** *For every integer $N \geq 0$,*

$$\mathsf{A}_k^E(N) = \mathrm{Cat}_k\, X^{k+1} + \sum_{r=2}^{k} \binom{2k}{k-r} X^{k-r} \mathsf{K}_r^E(N). \tag{6.7}$$

If

$$\mathsf{A}_E(z, X) = \sum_{k \geq 0} \mathsf{A}_k^E(N) z^k, \qquad \mathsf{K}_E(u, X) = \sum_{r \geq 2} \mathsf{K}_r^E(N) u^r,$$

and $U = zC(Xz)^2 = (C(Xz) - 1)/X$, then

$$\mathsf{A}_E(z, X) = XC(Xz) + \frac{\mathsf{K}_E(U, X)}{\sqrt{1 - 4Xz}}, \tag{6.8}$$

and

$$\mathsf{K}_E(u, X) = \frac{1 - Xu}{1 + Xu}\left[\mathsf{A}_E\left(\frac{u}{(1 + Xu)^2}, X\right) - X(1 + Xu)\right]. \tag{6.9}$$

Equivalently, for $r \geq 2$,

$$\mathsf{K}_r^E(N) = \sum_{k=2}^{r} (-1)^{r-k} q_{r,r-k} X^{r-k} \big(\mathsf{A}_k^E(N) - \mathrm{Cat}_k\, X^{k+1}\big). \tag{6.10}$$

Proof. The empty core is the interval $[0, 1]$; after k ear insertions it contributes X^{k+1}. For a nonempty r-pair core, Theorem 3.4 contributes X^{k-r}. Applying Theorem 6.1 gives (6.7). Summation and the same Catalan substitution used above give (6.8)–(6.9). □

7. The prefix-state cactus

Let w be the cyclic double-occurrence word of a pairing and set

$$s_\ell = O(\ell) \in Q_k := \{0, 1\}^k, \qquad 0 \leq \ell \leq 2k.$$

Then $s_0 = s_{2k} = \varnothing$, and consecutive states differ by toggling the coordinate written in the intervening slot. Keep the $2k$ traversed edge occurrences distinct, even when two of them have the same endpoints, and let $\Gamma(w)$ be the resulting support multigraph on the distinct visited states.

Lemma 7.1 (Cactus structure)**.** *The multigraph $\Gamma(w)$ is a cactus. Every block is a cycle of length $2r$ using r coordinate labels, each exactly twice; a doubled cube edge is regarded as a two-edge cycle with $r = 1$. Distinct blocks use disjoint coordinate sets.*

If $e(w)$ is the number of distinct exceptional states, then

$$e(w) = \sum_C (r_C - 1), \tag{7.1}$$

where the sum runs over the cycle blocks. If w is reduced, no two-edge block is a leaf block.

Proof. The word is an Euler circuit of $\Gamma(w)$, so the connected multigraph has no bridge. Suppose it contained a theta subgraph with distinct endpoints u, v. Choose a cube coordinate on which u and v differ. Each of the three edge-disjoint u-to-v paths would use that coordinate an odd number of times, requiring at least three occurrences of its label. Every label occurs exactly twice, a contradiction. A bridgeless block containing no theta is a cycle: otherwise, an ear attached to any cycle in the block produces a theta. Thus every block is a cycle.

A cube cycle uses each coordinate an even number of times. Since a label occurs only twice globally, a coordinate used by one block occurs twice there and nowhere else. Hence the blocks partition the k coordinates and $\sum_C r_C = k$. If there are c blocks, the cactus identity gives

$$|V(\Gamma)| = 2k - c + 1.$$

The empty state and the states immediately following the k first occurrences are $k + 1$ distinct basis states. All other visited states are exceptional, so

$$e(w) = |V(\Gamma)| - (k + 1) = k - c = \sum_C (r_C - 1).$$

Finally, the nonarticulation vertex of a leaf two-edge block has degree two. The Euler circuit must enter and leave it on the two equally labelled parallel edges, producing cyclically adjacent equal letters. Reducedness excludes this. □

Corollary 7.2 (Two-exception classification)**.** *Let w be reduced and suppose $e(w) = 2$. Then either $k = 3$, or, after cyclic shift, reversal, and relabelling,*

$$w = 1212\, p_1 \cdots p_\ell\, 3434\, p_\ell \cdots p_1, \qquad \ell = k - 4 \geq 0. \tag{7.2}$$

In the latter case the two exceptional-slab volume is $9/20$ if $\ell = 0$ and $4/9$ if $\ell > 0$.

Proof. Equation (7.1) leaves two possibilities: one six-cycle, or two square blocks, together with any two-edge blocks. In the first case, a nonempty tree of two-edge blocks attached to the six-cycle has an external leaf, contradicting reducedness. Thus $k = 3$. In the second case, the two square blocks are joined by a possibly empty chain of doubled edges and there can be no branch. Cutting an Euler circuit at an attachment vertex gives (7.2).

Write $X_j = \Phi_j - \frac{1}{2}$ for the centered opening coordinates. The first square contributes the exceptional normal

$$M_1 = X_0 - X_1 + X_2.$$

If the joining chain is nonempty, the state at the second square is the basis state created by its last opening p_ℓ, so the other exceptional normal is

$$M_2 = X_{p_\ell} - X_3 + X_4.$$

The supports are disjoint. For three independent centered unit uniforms,

$$\Pr\{|U - V + W| \leq \tfrac{1}{2}\} = \int_{-1/2}^{1/2} \left(\frac{3}{4} - x^2\right) dx = \frac{2}{3},$$

and hence the two-slab volume is $(2/3)^2 = 4/9$.

If the squares meet, then p_ℓ is absent and

$$M_2 = X_0 - X_3 + X_4.$$

Conditioning on the shared coordinate $X_0 = x$, either triangle slab has fiber probability $g(x) = 3/4 - x^2$. Therefore

$$\Pr\{|M_1| \le \tfrac{1}{2},\ |M_2| \le \tfrac{1}{2}\} = \int_{-1/2}^{1/2} g(x)^2\,dx = \frac{9}{20}.$$

This also identifies the two-square equality configuration without appealing to a finite pairing enumeration. □

Corollary 7.3 (Parity-mixing state count)**.** *In a parity-mixing word, every nontrivial block of $\Gamma(w)$ has $r_C \ge 3$. A reduced parity-mixing core with exactly three exceptional states has four pairs. Every reduced parity-mixing core with at least five pairs has at least four exceptional states.*

Proof. After suppressing the even-length excursions attached to a block, parity mixing is inherited by its cyclic direction word. A square has word *abab*, in which repeated labels occupy slots of the same parity; hence square blocks are forbidden. Thus $r_C \ge 3$ for every nontrivial block. If $e = 3$, (7.1) forces one eight-cycle, and reducedness rules out attached two-edge trees, so $k = 4$. For $k \ge 5$, the sum in (7.1) is at least four. □

8. Sharp slab inequalities

Translate the opening cube to $[-\frac{1}{2}, \frac{1}{2}]^{k+1}$. Since every slot form has coefficient sum one, a slot constraint becomes a centered unit slab

$$|M(X)| \le \frac{1}{2}. \tag{8.1}$$

The first exceptional form is always a triangle: if the initial opening run is $1, \dots, q$ and the first closing is $i < q$, then

$$L = \Phi_q - \Phi_i + \Phi_{i-1}. \tag{8.2}$$

Lemma 8.1 (Two-slab bound)**.** *Let $X_1, \dots, X_d$ be independent* Unif$[-1/2, 1/2]$ *variables, let*

$$L = X_1 + X_2 - X_3, \qquad M = \sum_{j=1}^{d} b_j X_j, \qquad b_j \in \mathbb{Z}, \qquad \sum_j b_j = 1,$$

and suppose M is neither L nor a coordinate. Then one of the following holds.

(a) *The form M is a triangle sharing exactly two coordinates with L, with the signs on the common coordinates simultaneously equal or simultaneously reversed. Then*

$$\Pr\{|L| \le 1/2,\ |M| \le 1/2\} = \frac{1}{2}.$$

(b) *Otherwise,*

$$\Pr\{|L| \le 1/2,\ |M| \le 1/2\} \le \frac{9}{20}. \tag{8.3}$$

Equality in (8.3) *occurs precisely, up to coordinate and sign symmetries, when M is a triangle sharing one coordinate with L, when $M = L + X_i - X_j$, or when $M = -L + X_i + X_j$, with distinct $i, j \ge 4$.*

Proof. The concentration reductions and the complete exact piecewise-polynomial integration are given in Appendix A. The argument treats arbitrary integral coefficients; no enumeration of pairings is used. □

Call the forms in alternative (a) *critical companions* of L.

Lemma 8.2 (Three critical slabs)**.** *If M and N are two distinct critical companions of L, then*

$$\Pr\{|L| \le 1/2,\ |M| \le 1/2,\ |N| \le 1/2\} \le \frac{2}{5}. \tag{8.4}$$

Proof. Every critical companion, for $j \ge 4$, is one of

$$\begin{aligned} A_j &= X_1 + X_2 - X_j, \quad B_j = X_1 - X_3 + X_j, \qquad C_j = -X_1 + X_3 + X_j, \\ D_j &= X_2 - X_3 + X_j, \quad E_j = -X_2 + X_3 + X_j. \end{aligned}$$

If the external coordinates agree, the exact values are $3/8$ or $17/48$. If they differ, the values are $11/30$ or $2/5$. These follow by conditioning on the common two-uniform sums; the full table appears in Appendix B. For example,

$$\Pr\{L, A_4, A_5 \text{ satisfy their slabs}\} = \int_{-1}^{1} (1 - |s|)^4 \, ds = \frac{2}{5}.$$

□

Lemma 8.3 (Lattice-point rigidity)**.** *Let $L = X_1 + X_2 - X_3$, and let M belong to one of the three noncritical equality families in Lemma 8.1. Then*

$$\operatorname{conv}\{e_1, \dots, e_d, L, M\} \cap \mathbb{Z}^d = \{e_1, \dots, e_d, L, M\}.$$

Proof. Let $G = [e_1 \ \cdots \ e_d \ L \ M]$. The matrix G is totally unimodular. Indeed, expansion along columns from the identity block reduces every minor to a minor of the two-column matrix $[L \ M]$. Its entries lie in $\{0, \pm 1\}$, and all its two-by-two minors lie in $\{0, \pm 1\}$. For the first equality family the supports of L and M meet in only one row. For $M = L + e_i - e_j$, the nonzero row types are (ℓ, ℓ), $(0, 1)$, and $(0, -1)$; for $M = -L + e_i + e_j$, they are $(\ell, -\ell)$ and $(0, 1)$, where $\ell \in \{\pm 1\}$.

Let z be an integral point of the displayed convex hull. Then $G\lambda = z$ for some $\lambda \ge 0$. Every column of G has coefficient sum one, so $\sum_i \lambda_i = \sum_i z_i = 1$. Total unimodularity and integrality of z imply that the nonempty bounded polytope

$$\{\lambda \ge 0 : G\lambda = z\}$$

has an integral vertex. Its coordinates are nonnegative integers summing to one, so it is a unit vector. Hence z is a column of G. □

9. The higher-core Toeplitz gap

Theorem 9.1 (Sharp higher-core gap)**.** *Let ρ be a reduced Toeplitz core with $k \ge 4$ pairs. Then*

$$p_T(\rho) \le \frac{9}{20}. \tag{9.1}$$

Equality holds if and only if $k = 4$ and, up to cyclic rotation, reflection, and relabelling of pairs,

$$\rho = 12123434. \tag{9.2}$$

Equivalently, its prefix-state cactus consists of two square blocks meeting in one vertex.

Proof. Let e be the number of distinct exceptional states. Reducedness rules out $e = 0$ and $e = 1$ here. If $e = 0$, Lemma 7.1 says that every block is a doubled edge; a leaf block then produces an ear. If $e = 1$, there is one square block and all remaining blocks are doubled edges. Any attached doubled-edge tree again has a leaf ear, so reducedness leaves only the square itself, which has $k = 2$. Consequently a reduced core with $k \geq 4$ has $e \geq 2$.

If $e = 2$, Corollary 7.2 applies. Since $k \geq 4$, the support consists of two square blocks joined by a doubled-edge chain. The volume is $9/20$ if the chain is empty and $4/9$ otherwise.

Suppose $e \geq 3$, and let L be the first exceptional triangle form from (8.2). If some other exceptional form is not a critical companion, Lemma 8.1 gives $p_T(\rho) \leq 9/20$. If every other exceptional form is critical, choose two of them and apply Lemma 8.2 to obtain $p_T(\rho) \leq 2/5$.

It remains to exclude equality when $e \geq 3$. Let (L, M) be a noncritical equality pair in Lemma 8.1, and put

$$K = \{x : |x_j| \leq \tfrac{1}{2},\ |L(x)| \leq \tfrac{1}{2},\ |M(x)| \leq \tfrac{1}{2}\}.$$

Since $0 \in \operatorname{int} K$,

$$K^\circ = 2\operatorname{conv}\{\pm e_1, \dots, \pm e_d, \pm L, \pm M\}. \tag{9.3}$$

Thus a further slab $|N(x)| \leq 1/2$ is redundant on K if and only if

$$N \in \operatorname{conv}\{\pm e_j, \pm L, \pm M\}.$$

Every exceptional normal is integral and has coefficient sum one. Applying that coefficient-sum functional to a convex representation of N forces the total weight on negative generators to vanish. By Lemma 8.3, N must then be a coordinate, L, or M, which is impossible for a third distinct exceptional state. A nonredundant slab removes positive volume: take a point of K violating it and move slightly toward the origin to obtain an interior point of K that still violates it. Equality is therefore impossible for $e \geq 3$. Hence equality forces $e = 2$, and Corollary 7.2 gives exactly (9.2). □

Corollary 9.2 (Quantized higher-core gap)**.** *If ρ is a reduced r-pair Toeplitz core with $r \geq 5$, then*

$$\frac{2^{s_2(r)}}{(r+1)!} \leq p_T(\rho) \leq \frac{9}{20} - \frac{2}{(r+1)!}.$$

The same bounds hold for every reduced parity-mixing Hankel core already for $r \geq 4$. Consequently, whenever the relevant core class is nonempty,

$$\frac{2^{s_2(r)}}{(r+1)!} \leq \frac{\kappa_r^E}{c_r^E} \leq \frac{9}{20} - \frac{2}{(r+1)!},$$

for $E = T,\ r \geq 5$, and for $E = H,\ r \geq 4$.

Proof. The lower bound is (5.2). Put $V = (r+1)!\, p_T(\rho)$. By Corollary 5.3, V is an even integer. The equality classification in Theorem 9.1 gives $p_T(\rho) < 9/20$ for $r \geq 5$, while $9(r+1)!/20$ is an even integer because $r + 1 \geq 6$. Thus

$$V \leq \frac{9}{20}(r+1)! - 2.$$

The Toeplitz equality core at $r = 4$ is not parity mixing, so the same strictness argument and Theorem 4.1 apply to Hankel cores from $r = 4$ onward. At the additional endpoint $r = 4$, the required parity is explicit: $9 \cdot 5!/20 = 54$ is even. Average the termwise bounds for the last assertion. □

Corollary 9.3 (All 1/2 extremizers)**.** *For a reduced Toeplitz core ρ,*

$$p_T(\rho) = \frac{1}{2} \quad \Longleftrightarrow \quad |\rho| = 3.$$

The four canonical reduced three-pair words are

$$121323, \qquad 123123, \qquad 123132, \qquad 123213.$$

For a reduced parity-mixing core,

$$p_H(\rho) = \frac{1}{2} \quad \Longleftrightarrow \quad \rho = 123123$$

up to the usual symmetries.

Proof. Each of the four three-pair words produces a critical two-triangle configuration and has volume 1/2. By Theorem 9.1, every larger reduced core has smaller volume. Among the four words, only 123123 is parity mixing; apply Theorem 4.1. □

Corollary 9.4 (Refined moment upper bounds)**.** *For every $k \geq 4$,*

$$\begin{aligned} m_{2k}(\gamma_T) &\leq \frac{9}{20}(2k-1)!! + \frac{11}{20}\operatorname{Cat}_k + \frac{13}{60}\binom{2k}{k-2} + \frac{1}{5}\binom{2k}{k-3} - 2\sum_{r=5}^{k}\binom{2k}{k-r}\frac{c_r^T}{(r+1)!}, \\ m_{2k}(\gamma_H) &\leq \frac{9}{20}k! + \frac{11}{20}\operatorname{Cat}_k + \frac{1}{20}\binom{2k}{k-3} - 2\sum_{r=4}^{k}\binom{2k}{k-r}\frac{c_r^H}{(r+1)!}. \end{aligned} \tag{9.4}$$

Here an empty sum is zero. Whenever nonempty, the displayed subtractions are universal positive improvements over the bare 9/20 higher-core bound; the Toeplitz subtraction begins at $k = 5$.

Proof. There are Cat_k empty-core pairings, $\binom{2k}{k-2}$ inflations of the unique two-pair core, and $4\binom{2k}{k-3}$ inflations of the four three-pair cores. Their respective volumes are 1, 2/3, and 1/2; all remaining pairings have volume at most 9/20. For each Toeplitz core of size $r \geq 5$, Corollary 9.2 improves this baseline by at least $2/(r+1)!$, and the number of its inflations is $\binom{2k}{k-r}$. Summing over the c_r^T reduced cores gives the first line. In the parity-mixing class there is no two-pair core and there is one three-pair core, 123123; the quantized improvement starts at $r = 4$, which gives the second line. □

10. The Hankel higher-core gap

The termwise equivalence in Theorem 4.1 converts the Hankel problem into a Toeplitz volume problem on the smaller parity-mixing class. The four-pair layer can be treated exactly.

Proposition 10.1 (The four-pair Hankel core)**.** *There are two labelled reduced parity-mixing four-pair words,*

$$12314324, \qquad 12342143,$$

and they lie in one dihedral orbit. Each has

$$p_H = p_T = \frac{11}{30}. \tag{10.1}$$

Proof. A parity-mixing pairing is a permutation between the even and odd slots. Reducedness excludes the two circular neighbours of each even slot. Inclusion–exclusion, or direct normalization of the two remaining permutations, gives the displayed words; reflection interchanges them.

For 12314324, the three exceptional forms can be chosen as

$$L_1 = X_0 - X_1 + X_3, \qquad L_2 = X_2 - X_3 + X_4, \qquad L_3 = X_1 - X_3 + X_4.$$

Put $a = X_3 - X_1$ and $b = X_4 - X_3$. Integrating first in X_0 and X_2, and then along the remaining X_1-fiber, gives

$$\begin{aligned} p_T &= \int_{\mathbb{R}^2} (1-|a|)_+(1-|b|)_+(1-|a|-|b|)_+ \, da\, db \\ &= 4\int_{\substack{a,b\geq 0 \\ a+b\leq 1}} (1-a)(1-b)(1-a-b)\, da\, db = \frac{11}{30}. \end{aligned}$$

Apply Theorem 4.1. □

Proposition 10.2 (Exact first four-exception layers)**.** *The* 13 *labelled reduced parity-mixing five-pair cores have volume distribution*

volume	1/3	13/45	49/180	1/4
multiplicity	1	5	5	2.

The unique 1/3 *core is* 1234512345*. At six pairs, the six labelled reduced parity-mixing cores with exactly four exceptional states form one dihedral orbit and all have volume* 23/90*. Over the complete reduced layers, the exact maxima at six and seven pairs are, respectively,*

$$\frac{23}{90} \qquad \textit{and} \qquad \frac{1}{4}.$$

Proof. Represent a parity-mixing pairing as a permutation from the even to the odd slots, reject the circular adjacencies, and quotient by rotations and reflections. This leaves four dihedral types at five pairs and one four-exception type at six pairs. For each representative, enumerate the rational vertices of its moment polytope, triangulate its boundary, and sum the simplex determinants over $\mathbb{Q}$. The complete reproducible calculation is `verify_hankel_layers.py`; it gives the displayed histogram and separately identifies the fully crossing representative. The complete exact orbit sweep underlying Theorem 11.1 contains all 80 reduced six-pair and 579 reduced seven-pair words and gives the two final maxima; the archived engine and file hashes are specified in the reproducibility statement. □

By Corollary 7.3, every reduced parity-mixing core with at least five pairs has at least four distinct exceptional states. We prove the needed joint concentration bound in a stronger totally-unimodular form.

For a rank-four real column matrix $D = (d_1, \ldots, d_n)$, let B_D denote the density, with respect to four-dimensional Lebesgue measure, of $\sum_i U_i d_i$, where the U_i are independent centered unit uniform variables. Thus B_D is the centered box spline associated with D.

Lemma 10.3 (Rank-four box-spline certificate)**.** *Let D be a rank-four totally unimodular matrix. Suppose every nonzero binary cocycle of the column matroid of D has even cardinality at least four. Then*

$$B_D(0) \leq \frac{1}{3}. \tag{10.2}$$

Proof. We give the reduction and the complete finite certificate. Zero columns do not change B_D. Column sign changes and unimodular row operations also preserve $B_D(0)$. Moreover, if D' is a spanning submultiset of the columns of D, then

$$B_D(0) \leq B_{D'}(0). \tag{10.3}$$

Indeed, a centered box spline is even and log-concave, by the Prékopa theorem [17] applied to the linear image of a cube, and hence is maximized at the origin. If $D = (D', d)$, adjoining the column d gives the one-column convolution formula

$$B_D(0) = \int_{-1/2}^{1/2} B_{D'}(-td)\,dt \le B_{D'}(0),$$

which proves (10.3); iterating handles an arbitrary spanning submultiset.

After simplification, the column matroid is a simple regular matroid of rank four. The rank-four classification says that it is isomorphic to a restriction of $M(K_5)$, except for the single additional type $M^*(K_{3,3})$ [7, 8]. To make the representation lift explicit, choose the same matroid basis in the two totally unimodular representations. Both basis matrices have determinant ± 1, so unimodular row operations normalize the representations to $[I_4 \mid A]$ and $[I_4 \mid A']$. Unique representability of regular matroids, equivalently uniqueness of their totally unimodular signings [15, Prop. 6.6.5], gives $A' = RAC$, componentwise, for diagonal sign matrices R, C; signs on the basis columns are absorbed into R. Undoing the normalizations therefore gives a row map in $\mathrm{GL}_4(\mathbb{Z})$ and individual column sign changes. Thus the equivalence uses only the operations noted above. It therefore suffices to use the ten directions of a reduced incidence matrix of K_5 and the nine directions of a fundamental cycle matrix of $K_{3,3}$.

Write m_e for the multiplicity of a frame direction e. For each of the fifteen nonzero binary row vectors u, put

$$w_u(m) = \sum_{u\cdot e = 1 \pmod 2} m_e.$$

The hypotheses are exactly that all $w_u(m)$ are even and at least four. If $m_e \ge 5$, subtracting two from m_e preserves these conditions: every affected weight was even and at least six. Hence it is enough to consider $m_e \in \{0, 1, 2, 3, 4\}$.

The remaining enumeration is exact. If valid $q < m$, then $s = (m - q) \bmod 2$ lies in the binary kernel of the frame. When $s \neq 0$, the valid vector $m - s$ still dominates q; when $s = 0$, one may subtract 2 in a coordinate on which $m - q$ is positive. Thus testing every admissible subtraction $s \le m$ by a nonzero binary kernel vector, and every $2e \le m$, finds every coordinatewise-minimal valid multiplicity. Modulo the frame automorphisms, there are 69 such vectors in the K_5 frame and 48 in the $M^*(K_{3,3})$ frame. Repeating a valid subtraction shows that every bounded valid multiplicity dominates one of these minima.

The accompanying exact certificate assigns to each of these 117 minima a spanning witness submultiset. There are 106 distinct witnesses. Twice applying (10.3), first from an arbitrary valid multiplicity to a dominated minimum and then to its witness, makes the witness volume an upper bound for the original one. Choosing a unimodular four-column basis converts each central box-spline value to the volume of a rational polytope. Exact vertex enumeration and determinant triangulation give maximum $1/3$, attained by two witnesses; the next-largest witness value is $211/640$. The script `verify_four_slabs.py` reconstructs both frames and their automorphism orbits, checks every reduction and containment relation, and recomputes all witness volumes over $\mathbb{Q}$. No floating-point decision enters the certificate. Equation (10.3) now proves the lemma. □

Theorem 10.4 (Four-slab theorem)**.** *Let $M \in \{0, \pm 1\}^{4\times d}$ have four distinct noncoordinate rows, each of coefficient sum one, and suppose M is totally unimodular. If X is uniform on $[-\frac{1}{2}, \frac{1}{2}]^d$, then*

$$\Pr\left\{|(MX)_j| \le \tfrac{1}{2},\ 1 \le j \le 4\right\} \le \frac{1}{3}. \tag{10.4}$$

In particular, this holds for any four distinct exceptional prefix-state forms of a parity-mixing double-occurrence word.

Proof. Set $D = [-M \mid I_4]$. By conditioning on the first d uniforms in the definition of the box spline,

$$B_D(0) = \int_{[-1/2,1/2]^d} \mathbf{1}_{[-1/2,1/2]^4}(Mx)\,dx = \Pr\left\{\|MX\|_\infty \le \tfrac{1}{2}\right\}.$$

The matrix D is rank four and totally unimodular. Since every row of M sums to one, $D\mathbf{1} = 0$. Hence, for every binary row vector u,

$$|\operatorname{supp}(uD)| \equiv uD\mathbf{1} = 0 \pmod 2,$$

so every binary cocycle has even cardinality.

A dual representation is $[I_d \mid M^{\mathsf{T}}]$. It is simple over $\mathbb{F}_2$. Indeed, a noncoordinate coefficient-sum-one row has odd support of size at least three. If two rows of M had the same binary support, the 2-by-2 minors would force their signs to be proportional on that support; coefficient sum one and distinctness exclude both possible proportionalities. Thus D is cosimple, so no nonzero cocycle has cardinality one or two. Every nonzero cocycle therefore has even cardinality at least four, and Lemma 10.3 applies. The final assertion follows from Lemma 2.5. □

The constant is sharp. For the fully crossing five-pair word 1234512345, the four exceptional centered normals are

$$M_j = X_0 + X_5 - X_j, \qquad 1 \le j \le 4.$$

Conditioning on $S = X_0 + X_5$, whose density is $(1 - |s|)_+$, each of the four remaining fibers has length $(1 - |s|)_+$. Hence their exact joint slab volume is

$$\int_{-1}^{1} (1 - |s|)^5\,ds = \frac{1}{3}.$$

Corollary 10.5 (Optimal Hankel gap)**.** *Every reduced parity-mixing core ρ with at least four pairs satisfies*

$$p_H(\rho) \le \frac{11}{30},$$

with equality precisely for the dihedral class in Proposition 10.1. For $|\rho| \ge 5$, one has $p_H(\rho) \le 1/3$.

Proof. The four-pair case is Proposition 10.1. A reduced parity-mixing core with at least five pairs has four distinct exceptional states by Corollary 7.3. Discard all other constraints, apply Theorem 10.4, and use Theorem 4.1. Since $1/3 < 11/30$, equality occurs only in the four-pair class. □

11. Exact moments

For each dihedral orbit representative, we deduplicate the defining halfspaces, enumerate vertices over $\mathbb{Q}$ by incremental double description, and triangulate using exact affine-rank tests. The volume is the sum of rational simplex determinants. At order twelve, the 10395 Toeplitz pairings reduce to 554 dihedral orbits. At order fourteen, 135135 Toeplitz pairings reduce to 5283 orbits; the Hankel class contains $7! = 5040$ pairings.

The implementation reproduces the Toeplitz moments through order six reported in [10, Sec. 2.4] and the Hankel moments through order eight recorded in [6, Prop. A.2]. At order eight, we additionally use a direct sum over all 105 Toeplitz pairings and a separate active-basis enumeration. The core transform, integral normalized-volume check from Theorem 5.1, and reconstruction from primitive masses provide exact internal consistency checks at every

order. The values through order fourteen first appeared in [20]; the computations here are independently organized around the core transform.

Theorem 11.1 (Exact moments through order fourteen)**.** *The even moments are*

$$\begin{array}{c|cc} 2k & m_{2k}(\gamma_T) & m_{2k}(\gamma_H) \\ \hline 2 & 1 & 1 \\ 4 & \frac{8}{3} & 2 \\ 6 & 11 & \frac{11}{2} \\ 8 & \frac{908}{15} & \frac{281}{15} \\ 10 & 415 & \frac{2717}{36} \\ 12 & \frac{23840}{7} & \frac{1052}{3} \\ 14 & \frac{325719}{10} & \frac{1331087}{720} \end{array} \tag{11.1}$$

In particular, the Toeplitz eighth moment is $908/15$*, rather than the value* $964/15$ *reported in* [10, Sec. 2.4].

The order-eight correction is transparent after sorting by core:

core type	number of pairings	volume
empty	14	1
two-pair core	28	2/3
three-pair cores	32	1/2
four-pair cores	4, 5, 22	9/20, 2/5, 11/30

Thus

$$14 + 28\frac{2}{3} + 32\frac{1}{2} + 4\frac{9}{20} + 5\frac{2}{5} + 22\frac{11}{30} = \frac{908}{15}.$$

The multiplicities sum to 105.

12. Reduced-core counts and Poisson peeling

Let

$$c_r^T = |\mathcal{R}_{T,r}|, \qquad c_r^H = |\mathcal{R}_{H,r}|.$$

Reduced Toeplitz cores are precisely the labelled loopless circular chord diagrams enumerated by Krasko and Omelchenko [13, Sec. 2, Thm. 2.2]. Reduced parity-mixing cores are equivalently ménage permutations, so their numbers are the classical ménage numbers and the second identity below is Touchard's inclusion–exclusion formula [21, 12]. We record both formulas in the present notation. Put

$$q_{r,j} = \frac{2r}{2r-j}\binom{2r-j}{j}.$$

Proposition 12.1 (Classical reduced-core counts)**.** *For* $r \geq 2$*,*

$$c_r^T = \sum_{j=0}^{r} (-1)^j q_{r,j} (2r-2j-1)!!, \qquad c_r^H = \sum_{j=0}^{r} (-1)^j q_{r,j} (r-j)!. \tag{12.1}$$

At the endpoints,

$$c_0^T = c_0^H = 1, \qquad c_1^T = c_1^H = 0.$$

For $r = 2, \ldots, 7$*,*

r	2	3	4	5	6	7
c_r^T	1	4	31	293	3326	44189
c_r^H	0	1	2	13	80	579

Proof. There are $2r$ forbidden circular edges. A set of j such edges can occur in a pairing only when they are disjoint, and there are $q_{r,j}$ choices. After forcing them, the remaining Toeplitz vertices have $(2r-2j-1)!!$ pairings. In the parity-mixing class, the remaining even vertices can be matched to the remaining odd vertices in $(r-j)!$ ways. Inclusion–exclusion proves (12.1). □

Choose a pairing uniformly from the relevant order-$2k$ class, and let D_k^E be the total number of pairs removed by complete core reduction. This includes cascading deletions, not only ears present initially.

Combining Flajolet and Noy's monolithic-diagram theorem and largest-component limit [9, Thms. 1–2] with the following observation implies the Pois(1) limit for ordinary complete peeling: on a monolith, every component outside the unique nontrivial component is a nonnested isolated edge removed by the ear cascade, while the nontrivial component has no ear, so $D_k^T = k - L_k$ in their notation. Paguyo [16, Thm. 1.2] gives an $O(k^{-1})$ approximation for ears present initially. Here we record the exact law of the cascading deletion statistic, recover its Poisson limit with an explicit rate, and obtain the parity-mixing analogue.

Theorem 12.2 (Poisson peeling)**.** *As $k \to \infty$,*

$$D_k^T \Rightarrow \mathrm{Pois}(1), \qquad D_k^H \Rightarrow \mathrm{Pois}(2). \tag{12.2}$$

Put $N_{T,k} = (2k-1)!!$ and $N_{H,k} = k!$. The full exact law is

$$\Pr(D_k^E = m) = \begin{cases} \dbinom{2k}{m} \dfrac{c_{k-m}^E}{N_{E,k}}, & 0 \le m \le k-2, \\ 0, & m = k-1, \\ \dfrac{\mathrm{Cat}_k}{N_{E,k}}, & m = k, \end{cases} \qquad E \in \{T, H\}. \tag{12.3}$$

For every fixed $m \ge 0$,

$$\Pr(D_k^T = m) \longrightarrow e^{-1}\frac{1}{m!}, \qquad \Pr(D_k^H = m) \longrightarrow e^{-2}\frac{2^m}{m!}. \tag{12.4}$$

More quantitatively, for $k \ge 3$,

$$d_{\mathrm{TV}}(D_k^T, \mathrm{Pois}(1)) + d_{\mathrm{TV}}(D_k^H, \mathrm{Pois}(2)) = O(k^{-1}). \tag{12.5}$$

Proof. For $0 \le m \le k-2$, (12.3) follows from Theorem 6.1 with $r = k-m$ and $F \equiv 1$. An empty core is the same thing as a noncrossing pairing, giving the $m = k$ line, while $D_k = k-1$ is impossible.

Let X_k^E count only the ears present before peeling. For Toeplitz pairings, [16, Theorem 1.2] gives

$$d_{\mathrm{TV}}(X_k^T, \mathrm{Pois}(1)) = O(k^{-1}).$$

For completeness, the parity-mixing analogue follows from the same size-bias argument. Index the even vertices by rows $i \in \mathbb{Z}/k\mathbb{Z}$ and the odd vertices by columns, and write a parity-mixing pairing as a uniform permutation σ. The two circular adjacencies in row i are the distinguished columns $\mathcal{D}_i = \{i, i-1\}$, so X_k^H is the number of occupied distinguished cells and has mean two.

Choose a distinguished cell (I, J) uniformly, put $L = \sigma(I)$ and $I' = \sigma^{-1}(J)$, and force (I, J) by swapping the images of I and I'. The resulting count $(X_k^H)^s$ has the X_k^H-size-biased law. Let $A = \{L = J\}$, let U be the event that L is the other member of $\mathcal{D}_I$, let V be the event that I' is the other row whose distinguished set contains J, and put $Z = \{A^c, L \in \mathcal{D}_{I'}\}$. Only the two swapped rows can change, and

$$|X_k^H + 1 - (X_k^H)^s| \le \mathbf{1}_A + \mathbf{1}_U + \mathbf{1}_V + \mathbf{1}_Z.$$

The first three expectations equal $1/k$, while $\Pr(Z) \le 2/k$. The size-bias Poisson inequality in [16, Theorem 3.1] therefore gives

$$d_{\mathrm{TV}}(X_k^H, \mathrm{Pois}(2)) \le \frac{5}{k}.$$

It remains to compare initial ears with the complete cascade. If $D_k^E \neq X_k^E$, some chord that was not initially an ear eventually becomes one. One of its two circular arcs therefore contains a nonempty pairing with empty core, hence a noncrossing pairing. If the chord together with that arc contains r pairs, there are Cat_{r-1} possibilities inside. A union bound over the $2k$ possible initial endpoints gives

$$\begin{aligned} \Pr(D_k^T \neq X_k^T) &\le B_T(k) = 2k \sum_{r=2}^{k-1} \frac{\mathrm{Cat}_{r-1}(2k-2r-1)!!}{(2k-1)!!}, \\ \Pr(D_k^H \neq X_k^H) &\le B_H(k) = 2k \sum_{r=2}^{k-1} \frac{\mathrm{Cat}_{r-1}(k-r)!}{k!}. \end{aligned} \tag{12.6}$$

where, after writing $s = r-1$,

$$B_H(k) = 2\sum_{s=1}^{k-2} \frac{\mathrm{Cat}_s}{(k-1)(k-2)\cdots(k-s)}, \qquad B_T(k) = 2k \sum_{s=1}^{k-2} \frac{\mathrm{Cat}_s}{\prod_{j=0}^{s}(2k-2j-1)} \le B_H(k).$$

Both are $O(k^{-1})$. Indeed, for $s \le k/2$, the bound $\mathrm{Cat}_s \le 4^s$ gives the geometric majorant $2\sum_{s\ge1}(8/k)^s$. For $s > k/2$, the descending product in $B_H(k)$ dominates $s!$, and the corresponding tail of $\sum_s 4^s/s!$ is $o(k^{-1})$. Coupling D_k^E and X_k^E on the same pairing proves (12.5).

Finally,

$$\frac{c_r^T}{(2r-1)!!} = \Pr(X_r^T = 0) \longrightarrow e^{-1}, \qquad \frac{c_r^H}{r!} = \Pr(X_r^H = 0) \longrightarrow e^{-2},$$

and, for fixed m,

$$\binom{2k}{m}\frac{(2k-2m-1)!!}{(2k-1)!!} \longrightarrow \frac{1}{m!}, \qquad \binom{2k}{m}\frac{(k-m)!}{k!} \longrightarrow \frac{2^m}{m!}.$$

This proves (12.4); since the limiting masses sum to one, it also proves (12.2). □

Proposition 12.3 (Sharp first-order peeling asymptotics)**.** *As $r \to \infty$,*

$$\frac{c_r^T}{(2r-1)!!} = e^{-1}\left(1 - \frac{3}{4r} + O(r^{-2})\right), \qquad \frac{c_r^H}{r!} = e^{-2}\left(1 - \frac{1}{r} + O(r^{-2})\right). \tag{12.7}$$

For every fixed $m \ge 0$,

$$\begin{aligned} \Pr(D_k^T = m) &= \frac{e^{-1}}{m!}\left[1 + \frac{m(m+1)-3}{4k} + O(k^{-2})\right], \\ \Pr(D_k^H = m) &= e^{-2}\frac{2^m}{m!}\left[1 + \frac{m(m-1)-4}{4k} + O(k^{-2})\right]. \end{aligned} \tag{12.8}$$

The total-variation constants are sharp:

$$\lim_{k\to\infty} k\, d_{\mathrm{TV}}(D_k^T, \mathrm{Pois}(1)) = e^{-1}, \qquad \lim_{k\to\infty} k\, d_{\mathrm{TV}}(D_k^H, \mathrm{Pois}(2)) = 4e^{-2}. \tag{12.9}$$

Moreover,

$$\begin{aligned}\mathbb{E}D_k^T &= 1+\frac{1}{k}+O(k^{-2}), \quad \operatorname{Var}(D_k^T) = 1+\frac{3}{2k}+O(k^{-2}),\\ \mathbb{E}D_k^H &= 2+\frac{2}{k}+O(k^{-2}), \quad \operatorname{Var}(D_k^H) = 2+\frac{4}{k}+O(k^{-2}).\end{aligned} \tag{12.10}$$

Proof. Uniformly for $0 \le j \le r^{1/3}$, the two inclusion–exclusion summands in (12.1), after division by the total class sizes, have the expansions

$$\begin{aligned}q_{r,j}\frac{(2r-2j-1)!!}{(2r-1)!!} &= \frac{1}{j!}\left[1+\frac{j(3-j)}{4r}+O\Big(\frac{(1+j)^4}{r^2}\Big)\right],\\ q_{r,j}\frac{(r-j)!}{r!} &= \frac{2^j}{j!}\left[1-\frac{j(j-1)}{4r}+O\Big(\frac{(1+j)^4}{r^2}\Big)\right].\end{aligned}$$

The remaining terms have a factorial majorant; summing the exponential series gives (12.7).

For fixed m, the remaining factors in (12.3) satisfy

$$\begin{aligned}\binom{2k}{m}\frac{(2k-2m-1)!!}{(2k-1)!!} &= \frac{1}{m!}\left[1+\frac{m(m+1)}{4k}+O_m(k^{-2})\right],\\ \binom{2k}{m}\frac{(k-m)!}{k!} &= \frac{2^m}{m!}\left[1+\frac{m(m-1)}{4k}+O_m(k^{-2})\right].\end{aligned}$$

Combining these with (12.7) proves (12.8). To justify summation, split at $m \le k^{1/8}$. On this range the displayed product expansions are uniform, with their remainders bounded by a constant times $(1+m)^8/k^2$ times the corresponding Poisson mass. For $0 \le m \le k-2$, the exact law and the crude bounds $c_r^T \le (2r-1)!!$, $c_r^H \le r!$ give

$$\Pr(D_k^T = m) \le \frac{1}{m!}\prod_{j=0}^{m-1}\frac{2k-j}{2k-2j-1}, \qquad \Pr(D_k^H = m) \le \frac{1}{m!}\prod_{j=0}^{m-1}\frac{2k-j}{k-j}.$$

Extend the two right-hand products algebraically through $m = k$. After division by $4^m/m!$, their successive ratios are respectively

$$\frac{2k-m}{4(2k-2m-1)}, \qquad \frac{2k-m}{4(k-m)}.$$

Both ratios increase with m, so each resulting sequence first decreases and then increases. The endpoint pairs for the Toeplitz and Hankel sequences are, respectively, $(1, 2^{-k})$ and $(1, 4^{-k}\binom{2k}{k})$, all at most one. The $m = k-1$ atom is zero, while

$$\Pr(D_k^T = k) = \frac{2^k}{(k+1)k!}, \qquad \Pr(D_k^H = k) = \frac{\binom{2k}{k}}{(k+1)k!},$$

so these atoms satisfy the same bound. Thus, uniformly in k, m,

$$\Pr(D_k^E = m) \le \frac{4^m}{m!}.$$

Stirling's formula now gives, for every fixed $q, A \ge 0$,

$$k^A \sum_{m > k^{1/8}} m^q \frac{4^m}{m!} = o(1).$$

Dominated summation is therefore valid in ℓ^1 and in the weighted ℓ^1 spaces used below. In the Toeplitz case the first-order correction is negative only at $m = 0, 1$, whose total negative mass is e^{-1}. In the Hankel case it is negative at $m = 0, 1, 2$, whose total negative mass is $4e^{-2}$. This

proves (12.9). Finally, summing m and m^2 against the two first-order Poisson corrections gives (12.10). □

13. Infinite-divisibility obstructions

Proposition 13.1 (Classical infinite divisibility)**.** *Neither γ_T nor γ_H is classically infinitely divisible.*

Proof. Both laws are centered and have variance one. Their fourth classical cumulants are

$$\operatorname{cum}_4(\gamma_T) = \frac{8}{3} - 3 = -\frac{1}{3}, \qquad \operatorname{cum}_4(\gamma_H) = 2 - 3 = -1.$$

For a classically infinitely divisible law with finite fourth moment, the Lévy–Khintchine representation gives

$$\operatorname{cum}_4 = \int_{\mathbb{R}} x^4 \, \nu(dx) \geq 0,$$

where ν is the Lévy measure; the Gaussian component contributes only to the second cumulant. This is impossible; see [11]. □

Proposition 13.2 (Free infinite divisibility of the Hankel law)**.** *The Hankel limit γ_H is not freely infinitely divisible.*

Proof. Let (r_j) be its free cumulants. The noncrossing moment–cumulant recursion and

$$m_2 = 1, \quad m_4 = 2, \quad m_6 = \frac{11}{2}, \quad m_8 = \frac{281}{15}$$

give

$$r_2 = 1, \qquad r_4 = 0, \qquad r_6 = \frac{1}{2}, \qquad r_8 = \frac{11}{15}.$$

For a freely infinitely divisible law with moments of all orders, its free generating measure σ has moments of all orders [2, Proposition 2.3]. The free Lévy–Khintchine formula then gives

$$r_{n+2} = \int_{\mathbb{R}} x^n (1 + x^2) \, \sigma(dx), \qquad n \geq 0,$$

so every finite principal submatrix of $(r_{i+j})_{i,j \geq 1}$ is positive semidefinite; compare [14, Theorem 13.16] for the compactly supported formulation. The block indexed by 2 and 4 is

$$\begin{pmatrix} r_4 & r_6 \\ r_6 & r_8 \end{pmatrix} = \begin{pmatrix} 0 & 1/2 \\ 1/2 & 11/15 \end{pmatrix},$$

whose determinant is $-1/4$. □

The corresponding Toeplitz tests do not yet give an obstruction. The even free cumulants through order fourteen are

$$\begin{aligned} &r_2 = 1, \qquad r_4 = \frac{2}{3}, \qquad r_6 = 2, \qquad r_8 = \frac{454}{45}, \\ &r_{10} = \frac{619}{9}, \quad r_{12} = \frac{546794}{945}, \quad r_{14} = \frac{1546397}{270}. \end{aligned} \tag{13.1}$$

After permuting indices by parity, all conditional-positivity blocks accessible from these values are positive definite. The leading principal minors of the odd block indexed by $1, 3, 5, 7$ are

$$1, \quad \frac{14}{9}, \quad \frac{48814}{2025}, \quad \frac{226121391682}{28704375},$$

and those of the even block indexed by $2, 4, 6$ are

$$\frac{2}{3}, \quad \frac{368}{135}, \quad \frac{109932862}{637875}.$$

At order sixteen, write $A = (r_{i+j})_{i,j\in\{2,4,6\}}$ and

$$v = (r_{10}, r_{12}, r_{14})^{\mathsf{T}}.$$

Moment–cumulant inversion and exact elimination give

$$r_{16} = m_{16} - \frac{196536946}{675}, \qquad v^{\mathsf{T}} A^{-1} v = \frac{137490819049597003}{2181617646390}.$$

Since A is positive definite, the Schur-complement condition for the enlarged even block reduces exactly to

$$m_{16}(\gamma_T) \geq \frac{3863516832738344459}{10908088231950} \approx 354188.2638. \tag{13.2}$$

Thus an exact sixteenth moment below (13.2) would disprove free infinite divisibility; a value at or above it would merely pass every conditional-positivity test through order sixteen.

14. What is solved and what remains open

The core method now gives a complete all-order Toeplitz hierarchy through the four-pair layer: volume $2/3$ belongs exactly to inflations of the unique two-pair crossing, volume $1/2$ belongs exactly to inflations of reduced three-pair cores, and every larger core has volume at most $9/20$, with a unique reduced equality type. It also shows that all moment polytopes are index-two Gorenstein lattice polytopes and that Hankel contributions coincide termwise with Toeplitz contributions on the parity-mixing class. In the Hankel subclass, Theorem 10.4 and Corollary 10.5 give the sharp all-order hierarchy: the unique reduced four-pair class has volume $11/30$, and every larger reduced core has volume at most $1/3$.

Two larger identification problems remain.

(1) *Closed form and joint $*$-laws.* Identify either limiting scalar law explicitly, or give structural descriptions of the joint $*$-distributions of the Bose–Vishwakarma pairs Z_1, Z_2 and Y_1, Y_2. Their mixed moments already admit finite-integral descriptions [4, Thms. 4.3–4.4]; what remains open is a recognizable closed-form or standard dependence model. The identities $p_H(\pi) = p_T(\pi)$ for parity-mixing pairings give all-order scalar-moment consistency checks for $(Y_1 + Y_2)/\sqrt{2}$, but do not by themselves determine the mixed $*$-moments.

(2) *Primitive-volume asymptotics.* Determine the scale of

$$\frac{\kappa_r^E}{c_r^E}.$$

The exact values through seven pairs decay, but do not yet distinguish exponential decay from a more complicated scale. In particular, no estimate depending only exponentially on core size can hold pointwise without additional structural hypotheses: separated cactus blocks generate persistent-volume families.

We do not know whether γ_T is freely infinitely divisible. Within the conditional-positivity test used here, (13.2) makes $m_{16}(\gamma_T)$ the next undetermined finite target.

Reproducibility

The accompanying source package contains eight checkers with distinct roles.

- `verify_structural.py` checks algebraic consistency of the displayed counts, transforms, cumulants, threshold, sharp Poisson coefficients, and cascade-shell bounds.
- `verify_order8.py` independently constructs all 105 order-eight pairing polytopes and recomputes the exact 908/15 histogram from Ehrhart counts.
- `verify_slabs.py` regenerates the 9/19/26 admissible row lists, enumerates their rational vertices, and recomputes both sharp-slab tables with exact determinants; SciPy is used only to obtain a boundary triangulation.
- `verify_hankel_layers.py` exactly checks the four-pair endpoint, the five-pair Hankel volume distribution, and the six-pair four-exception orbit.
- `verify_rational_certificates.py` independently recomputes all two-slab rows and the first two four-exception Hankel layers using only the included pure-rational double-description and face-triangulation engine; no floating hull volume enters these values.
- `verify_four_slabs.py` reconstructs the two rank-four regular matroid frames, checks signed-unimodular lifts of every frame automorphism, enumerates the $69 + 48$ minimal multiplicity orbits, verifies their spanning witness cover, and recomputes the 106 distinct terminal box-spline volumes over $\mathbb{Q}$.
- `verify_archived_data.py` verifies the frozen exact-data hashes, reconstructs both order-fourteen moments, and checks the complete Hankel core counts and maximum volumes through seven pairs.
- `verify_all_moments.py` performs an optional, separate lattice-normalized orbit sweep through order fourteen.

The rank-four verifier uses exact bounded integer arrays and the included pure-rational volume engine; no floating-point decision enters the proof. The appendices record the complete finite slab tables.

The source package includes a frozen copy of the exact rational orbit-volume engine and the order-fourteen machine-readable data underlying (11.1). The complete source package for this revision is maintained at

`https://github.com/yspennstate/crossing-cores-rmt`.

The order-fourteen engine and data were imported from the independently audited legacy snapshot commit `2e0eafce`; the four-slab certificate and its verifier are additions in the present package. A frozen release, preferably with a DOI, together with file hashes and clean-recomputation instructions, should accompany submission.

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Declaration on generative AI and AI-assisted technologies

During manuscript preparation, generative-AI tools were used for language editing, organizational assistance, finite-case exploration, and algebraic cross-checking. Exact scripts and the cited sources provide the checks specified in the reproducibility section; responsibility for final verification and for the manuscript remains with the author.

Declaration of competing interest

The author declares no competing interests.

Appendix A. The exact two-slab calculation

We prove Lemma 8.1. Write $Z_1 = X_1$, $Z_2 = X_2$, and $Z_3 = -X_3$, so that $L = Z_1 + Z_2 + Z_3$, and put $c = (b_1, b_2, -b_3)$. Call coefficients with index at least four external.

If an external coefficient has absolute value at least two, conditioning on that variable gives a fiber of length at most $1/2$, independently of the L-event. The joint probability is at most $(1/2)(2/3) = 1/3$. If at least three external coefficients are nonzero and all are ± 1, the concentration of any selected three-uniform sum on a unit interval is at most $2/3$. Conditioning on the remaining variables gives at most $(2/3)^2 = 4/9$.

It remains to consider $r = 0, 1, 2$ external coefficients, all ± 1. If $|c_i - c_j| \geq 3$, condition on every variable except Z_i, Z_j. The two slab normals in these variables form the matrix

$$\begin{pmatrix} 1 & 1 \\ c_i & c_j \end{pmatrix},$$

so the inverse image has area at most $1/|c_i - c_j| \leq 1/3$. We may therefore assume $\max c_i - \min c_i \leq 2$.

Let

$$h_0(t) = \mathbf{1}_{\{|t| \leq 1/2\}},$$
$$h_1(t) = (1 - |t|)_+,$$
$$h_2(t) = \begin{cases} 3/4 - t^2, & |t| \leq 1/2, \\ \frac{1}{2}(3/2 - |t|)^2, & 1/2 \leq |t| \leq 3/2, \\ 0, & |t| \geq 3/2. \end{cases}$$

After harmless sign changes of external variables, the desired probability is

$$I_r(c) = \int_{[-1/2,1/2]^3} \mathbf{1}_{\{|z_1+z_2+z_3| \leq 1/2\}} h_r(c_1 z_1 + c_2 z_2 + c_3 z_3)\, dz. \tag{A.1}$$

Let $m = \min_i c_i$ and write $c_i = m + \delta_i$, where $\delta_i \in \{0, 1, 2\}$ and $\min_i \delta_i = 0$. If E is the sum of the external signs, then the coefficient-sum condition is

$$m = 1 - E - \delta_1 - \delta_2 + \delta_3. \tag{A.2}$$

For $r = 0, 1, 2$, respectively, E lies in

$$\{0\}, \qquad \{-1, 1\}, \qquad \{-2, 0, 2\}.$$

Equations (A.1)–(A.2), after permutation of the three Z-coordinates and removal of the coordinate and L cases, leave exactly the 9, 19, and 26 rows below. Integrating first in z_3 gives rational piecewise polynomials; the break hyperplanes are $z_1 + z_2 + z_3 = \pm 1/2$ and

$$c_1 z_1 + c_2 z_2 + c_3 z_3 \in \{\pm 1/2, \pm 1, \pm 3/2\}.$$

r	$\{c_1,c_2,c_3\}$	$I_r(c)$	$\{c_1,c_2,c_3\}$	$I_r(c)$
0	$(-3,-1,-1)$	11/36	$(-2,-1,0)$	5/12
	$(-1,-1,1)$	5/12	$(0,1,2)$	5/12
	$(1,2,2)$	5/12	$(1,3,3)$	8/27
	$(2,2,3)$	11/36	$(2,3,4)$	17/72
	$(3,3,5)$	26/135		
1	$(-4,-2,-2)$	23/96	$(-3,-2,-1)$	11/36
	$(-2,-2,0)$	17/48	$(-2,-1,-1)$	5/12
	$(-2,0,0)$	17/48	$(-1,-1,0)$	1/2
	$(-1,0,1)$	5/12	$(0,0,2)$	17/48
	$(0,1,1)$	1/2	$(0,2,2)$	17/48
	$(1,1,2)$	5/12	$(1,2,3)$	11/36
	$(2,2,2)$	17/48	$(2,2,4)$	23/96
	$(2,3,3)$	29/108	$(2,4,4)$	41/192
	$(3,3,4)$	47/216	$(3,4,5)$	13/72
	$(4,4,6)$	89/576		
2	$(-5,-3,-3)$	407/2160	$(-4,-3,-2)$	41/180
	$(-3,-3,-1)$	583/2160	$(-3,-2,-2)$	23/80
	$(-3,-1,-1)$	23/80	$(-2,-2,-1)$	17/48
	$(-2,-1,0)$	17/48	$(-1,-1,-1)$	9/20
	$(-1,-1,1)$	17/48	$(-1,0,0)$	9/20
	$(-1,1,1)$	17/48	$(0,0,1)$	9/20
	$(0,1,2)$	17/48	$(1,1,1)$	9/20
	$(1,1,3)$	23/80	$(1,2,2)$	17/48
	$(1,3,3)$	583/2160	$(2,2,3)$	23/80
	$(2,3,4)$	41/180	$(3,3,3)$	13/54
	$(3,3,5)$	407/2160	$(3,4,4)$	19/96
	$(3,5,5)$	1/6	$(4,4,5)$	27/160
	$(4,5,6)$	7/48	$(5,5,7)$	9/70

The two 1/2 rows, $(-1,-1,0)$ and $(0,1,1)$ with one external variable, are the two sign patterns for a triangle sharing two coordinates with L; these are exactly the critical companions. With two external variables, the 9/20 rows $(-1,0,0)$ and $(0,0,1)$ give the two sign patterns for a triangle sharing one coordinate with L, the row $(1,1,1)$ gives $M = L + X_i - X_j$, and $(-1,-1,-1)$ gives $M = -L + X_i + X_j$. Thus the four rows collapse under cube symmetries to the three noncritical equality families in Lemma 8.1. Every other entry is smaller. The accompanying `verify_slabs.py` reconstructs every rational vertex and recomputes all 54 volumes by exact determinants. This proves the lemma.

Appendix B. The three-critical-slab table

Use the notation $A_j, \dots, E_j$ from the proof of Lemma 8.2. Conditioning on the common sums gives the following exact values.

same external	A	B	C	D	E
A	–	3/8	3/8	3/8	3/8
B		–	17/48	3/8	3/8
C			–	3/8	3/8
D				–	17/48
E					–

different externals	A	B	C	D	E
A	2/5	11/30	11/30	11/30	11/30
B	11/30	2/5	2/5	11/30	11/30
C	11/30	2/5	2/5	11/30	11/30
D	11/30	11/30	11/30	2/5	2/5
E	11/30	11/30	11/30	2/5	2/5

The largest entry is 2/5, proving Lemma 8.2.

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